

BUSINESS REQUIREMENTS SPECIFICATION (BRS)

1. Document Control

Field	Details
Document Title	Business Requirements Specification (BRS)
Project Name	Virtual Worker Platform
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2. Executive Summary

2.1 Problem Statement

Organizations currently handle a large volume of repetitive digital tasks manually, including drafting communications, updating systems, generating reports, scheduling meetings, and processing structured operational requests. These tasks are time-consuming, error-prone, and dependent on human intervention. Existing automation tools require technical expertise and lack intelligent interpretation capabilities. There is no centralized, governed system that combines AI-driven task understanding with controlled enterprise workflow execution.

2.2 Proposed Solution (High-Level)

The Virtual Worker Platform will provide a governed AI-powered task execution system that accepts natural language requests, interprets them using Hugging Face AI models, maps them to controlled workflow templates, and executes them via n8n automation workflows. The system will include approval gates, audit logging, multi-tenant isolation, and role-based governance controls.

2.3 Project Vision / Future State

The future state envisions a scalable digital workforce layer where users can submit tasks conversationally, and the system executes them securely and reliably across integrated enterprise systems. The platform will reduce operational friction, standardize processes, and enable safe AI-assisted automation at scale.

2.4 Expected Business Value

- Reduction in manual operational workload by at least 50%
 - Increased process consistency and reduced human error
 - Improved compliance through enforced approvals and audit trails
 - Faster task turnaround times
 - Foundation for scalable enterprise AI workforce deployment
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3. Background & Business Context

3.1 Current Situation Overview

Current operations rely heavily on manual task execution or isolated automation scripts. AI tools exist but are disconnected from execution systems and lack governance controls. Workflow automation platforms exist but require technical configuration and do not interpret natural language tasks.

3.2 Business Drivers

- Digital transformation initiatives
- Operational cost reduction
- Increased productivity targets
- Risk mitigation and governance requirements
- Competitive advantage through intelligent automation

3.3 Related Projects / Systems

- Existing workflow automation tools
 - Enterprise email and calendar systems
 - CRM systems
 - Document management systems
 - Internal databases
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4. Business Goals & Strategic Objectives

Objective ID	Objective Description	Measurement Method	Target
OBJ-01	Reduce manual task workload	Time tracking analysis	50% reduction
OBJ-02	Increase automation rate	% of tasks executed automatically	70%+

Objective ID	Objective Description	Measurement Method	Target
OBJ-03	Improve compliance	Audit review results	100% traceable actions
OBJ-04	Improve task turnaround time	Avg completion time	30% faster

5. Scope Definition

5.1 In-Scope

- Multi-tenant user and workspace management
- Role-Based Access Control (RBAC)
- Natural language task submission (UI & API)
- AI classification and entity extraction
- Template-based workflow mapping
- n8n workflow execution integration
- Approval workflows for sensitive tasks
- Audit logging and reporting
- Notifications (Email/Slack)

5.2 Out-of-Scope

- Fully autonomous AI without governance controls
- Robotic desktop automation (RPA)
- Financial transactions without approval mechanisms
- Unrestricted external system modification

5.3 Scope Assumptions

- APIs for integrated systems are available
- Users have internet connectivity
- n8n and Hugging Face services are operational

6. Stakeholders & Governance

6.1 Stakeholder Register

Stakeholder	Role	Department	Responsibility
Product Owner	Sponsor	Strategy	Vision & funding
Technical Lead	Lead Architect	IT	Technical oversight
Automation Engineer	Workflow Developer	IT	n8n implementation
AI Engineer	AI Integration	IT	HF model integration
Security Lead	Compliance	Risk	Security validation
End Users	Operational Users	Various	Task submission
Approvers	Managers	Operations	Approval decisions

6.2 User Personas

Persona	Description	Key Needs
Admin	System administrator	Control, governance, visibility
End User	Daily operational user	Ease of use, speed
Approver	Managerial role	Oversight, risk control

7. Business Process Overview

7.1 Current State (As-Is)

Tasks are manually created, executed, and tracked across multiple systems. There is limited automation and no centralized AI-driven task interpretation.

7.2 Future State (To-Be)

Users submit tasks in natural language. The system interprets intent, matches it to approved templates, executes workflows via n8n, enforces approval gates, and logs all actions.

7.3 Process Gaps Identified

- No centralized task automation layer

- No AI interpretation linked to execution
- Limited governance over automation scripts
- Lack of full audit traceability

8. Business Requirements

8.1 Functional Requirements (FR)

Req ID	Requirement Description	Priority	Owner	Acceptance Criteria
FR-01	Users can submit tasks via UI/API	High	Backend Lead	Task stored successfully
FR-02	System classifies tasks using AI	High	AI Engineer	Correct template matched >80%
FR-03	System extracts structured inputs	High	AI Engineer	JSON inputs populated
FR-04	System triggers n8n workflows	High	Automation Engineer	Workflow execution ID captured
FR-05	Approval gates pause sensitive tasks	High	Backend Lead	Workflow resumes after approval
FR-06	System logs all actions	High	Security Lead	Audit entries recorded

9. Non-Functional Requirements (NFR)

9.1 Performance

Req ID	Requirement	Target
NFR-01	AI interpretation time	< 5 seconds

9.2 Security

Req ID	Requirement	Target
NFR-02	Multi-factor authentication	Required for Admin

9.3 Availability & Reliability

Req ID	Requirement	Target
NFR-03	System uptime	99.5% monthly

9.4 Scalability

Req ID	Requirement	Target
NFR-04	Concurrent users supported	1,000+

9.5 Compliance

Req ID	Requirement	Target
NFR-05	Data protection compliance	GDPR / Local Regulation

9.6 Audit & Logging

Req ID	Requirement	Target
NFR-06	Log admin actions	Full audit trail

9.7 Disaster Recovery

Req ID	Requirement	Target
NFR-07	Recovery Time Objective (RTO)	4 hours
NFR-08	Recovery Point Objective (RPO)	15 minutes

10. Business Rules

Rule ID	Business Rule	Applies To
BR-01	External communication requires approval	Email workflows
BR-02	All workflows must map to approved template	Task execution

11. Data Requirements

11.1 Data Inputs

Data Item	Source	Format	Frequency
Task Description	User	Text	Per Task
Attachments	User	PDF/DOCX/IMG	Per Task

11.2 Reports & Outputs

Report Name	Audience	Frequency
Task Performance Dashboard	Admin	Weekly

11.3 Data Retention

Data Type	Retention Period	Reason
Task Logs	7 Years	Compliance

12. Integration & External Interfaces

12.1 Systems for Integration

System Name	Purpose	Integration Type
Email Server	Send emails	API
Calendar System	Schedule meetings	API
CRM	Update records	API

12.2 Interface Requirements

All integrations must use secure API authentication and encrypted communication.

13. Reporting & Analytics Requirements

13.1 Dashboards

Dashboard	Users	Metrics
Operations Dashboard	Admin	Success rate, execution time

13.2 Key KPIs

- Workflow success rate
 - Automation rate
 - Approval turnaround time
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14. Constraints, Assumptions & Dependencies

14.1 Constraints

- Budget limits
- Fixed rollout timeline
- Regulatory compliance

14.2 Assumptions

- Required APIs are available
- n8n supports required integrations

14.3 Dependencies

Dependency	Owner	Risk if Not Met
API Access	IT	Integration delays

15. Risks & Mitigation Plan

Risk ID	Risk Description	Probability	Impact	Mitigation
R-01	AI misclassification	Medium	High	Confidence threshold
R-02	Workflow failure	Medium	High	Retry + alerting

16. Implementation Approach

16.1 Rollout Strategy

Phased rollout starting with pilot group.

16.2 Required Environments

Environment	Purpose
Development	Build
UAT	Testing
Production	Go-live

17. Glossary of Terms

Term	Definition
BRS	Business Requirements Specification
n8n	Workflow automation platform
HF	Hugging Face
RBAC	Role-Based Access Control

18. Appendices

Appendix A: As-Is Process Diagram (To be attached) Appendix B: To-Be Process Diagram (To be attached)
Appendix C: Wireframes / Mockups (To be attached)

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